SMART BUS SYSTEM FOR REAL TIME PASSENGER MANAGEMENT SYSTEM

SLIIT
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Meet our team

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Research Problem

- Existing bus tracking systems provide **limited real-time information** and do not integrate:
  - Passenger crowd monitoring
  - Accurate ETA prediction
  - Passenger–driver communication
- Passengers face:
  - Unpredictable arrival times
  - Severe overcrowding
  - Lack of real-time information
- Transport authorities lack data-driven decision systems.

Impact

- Long waiting times
- Passenger discomfort and safety issues
- Inefficient bus fleet management

Research Problem & Stakeholders

- Public bus transportation in Sri Lanka serves over 1.45 million passengers daily but lacks modern digital management systems.

Stakeholders

Primary Users :

- Daily commuters
- Travelers unfamiliar with routes

Secondary Users :

- Bus drivers
- Transport authorities (SLTB)

Decision Makers :

- Government transport authorities
- Private bus operators

Solution Overview

A Smart Bus Management System that provides real-time bus tracking, passenger information, and fleet management.

Key Features :

1. IoT Bus Tracking & ETA Prediction
2. Fleet Management & Route Optimization
3. Mobile Passenger Interaction System
4. Vision-based passenger counting

Non- Functional Focus

- Usability (simple mobile interface)
- Reliability (real-time data updates)
- Scalability (support multiple routes)

System Architecture and Integration

Main Components

1. IoT based Bus Tracking System & ETA Prediction
   Owner: IT22131560 – Weerakoon W.M.D.P

2. Fleet Management & Route Optimization
   Owner: IT22230492 – Pathiraja H.P.M.O.N

3. Mobile Passenger Interaction System
   Owner:IT22286246 – Chandrasekara C.M.P.V

4. Vision-based passenger counting & Occupancy
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Integration

- GPS & sensor data → MongoDB database
- APIs connect backend services to mobile application
- Camera data processed for passenger counting

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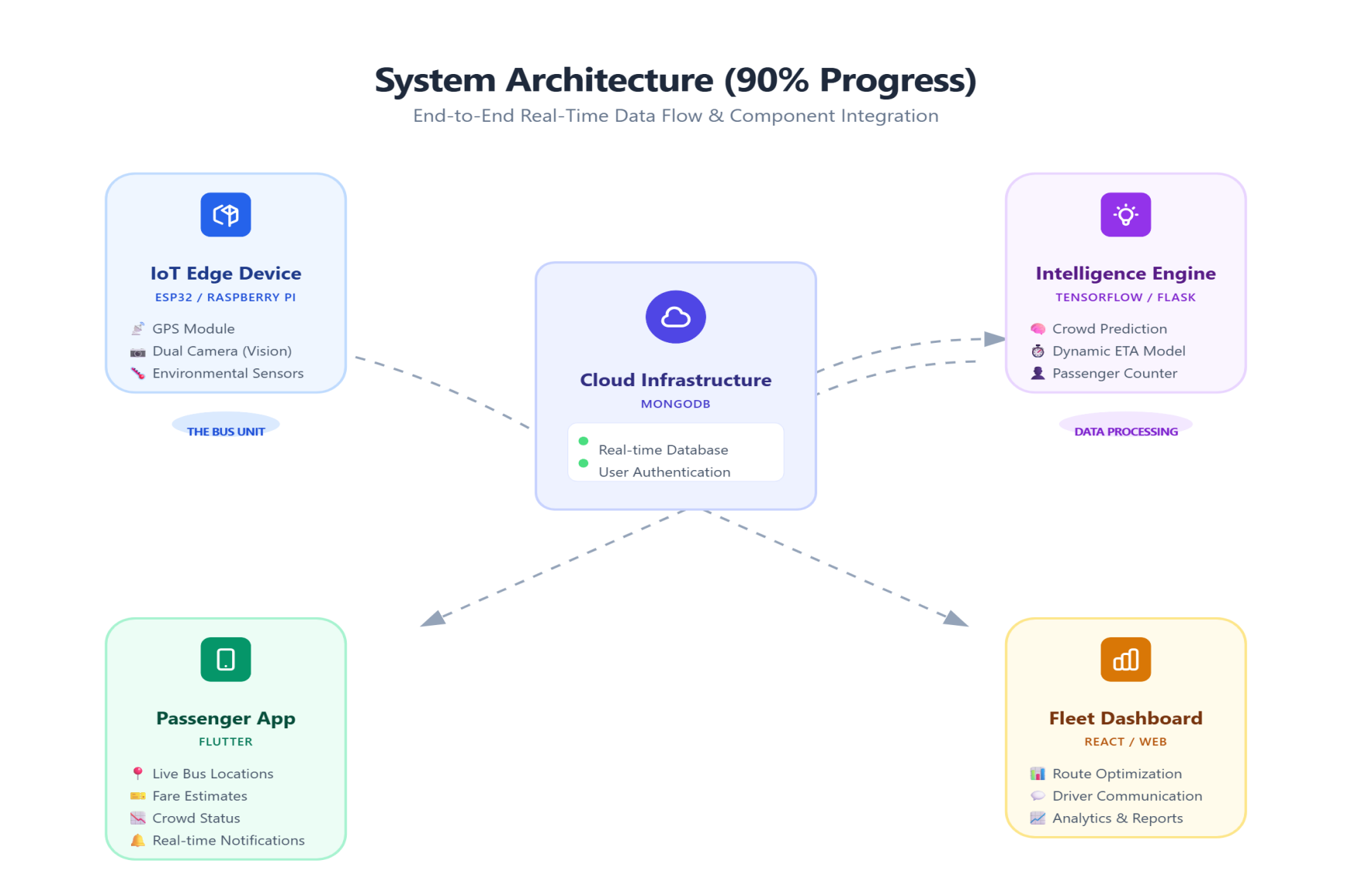

Commercialization

Target Market

- SLTB buses
- Private bus operators
- Daily commuters

Market Size

- Approximately 20,000 buses are available for public transportation in Sri Lanka.
- 4 – 5 million passengers use public transport buses daily.

Value Proposition

- Real-time bus information
- Reduced waiting time
- Improved passenger safety
- Data-driven fleet management

Thank You!